



CentraleSupélec

CENTRALESUPÉLEC
AUTOMATIC CONTROL DEPARTMENT

Laboratory work - Control Theory course

Modeling and control of an overhead crane



2023-2024

Preliminary remark: this study includes work to be done, outside the class-hours, between the two sessions scheduled in the timetable. It is essential to treat this part seriously in order to complete the work requested during the second session. The preparation will be checked individually by the supervisors at the beginning of the second session.

The report, which must covers both sessions as well as the preparation between the two sessions, must be returned 7 days after the end of the last session. It must be submitted in a PDF file through your cohort EDUNAO page.

1 Context and system description

An overhead crane is commonly used in various industrial activities in the transport for the loading and unloading of freight (for instance: loading containers on ships or loading of the fuel rods in nuclear power plants). The displacement of the load, placed at the end of a cable, is assured by a cart that moves on one rail (or two rails), one may see Figure 1. The cart is driven by a dc motor on the horizontal way and, the lift of the load is assured by another dc motor on the vertical way through a winch system.

The issues to be considered in controlling an overhead crane are to determine a control law in order to drive the cart from an initial point to a given final one without any oscillation. Load oscillations limit the cycle time in overhead crane transport operations and pose safety problems for people in and around the crane environment.

The same type of problem also arises for certain means of transport such as the telepheric cable car, see for example Figure 2.



Figure 1: Pont roulant et applications



Figure 2: Transport au moyen de téléphérique

The subject of this study is the design and validation of a control law on an experimental setup presenting the main characteristics of the system under consideration. The case of study

is limited to the only displacement on a horizontal axis, the length of the pendulum is then constant.

1.1 Experimental setup description

The support (Figure 3) illustrating the study is based on a cart driven using a system of cables and pulleys, and by a motorization system that consists of

- a DC machine with permanent magnets associated with a gearbox;
- a power amplifier to supply the electric machine;
- two position sensors to measure the longitudinal displacement of the cart and the angular position of the pendulum.

The cart supports a pendulum of constant length to which the load is attached.

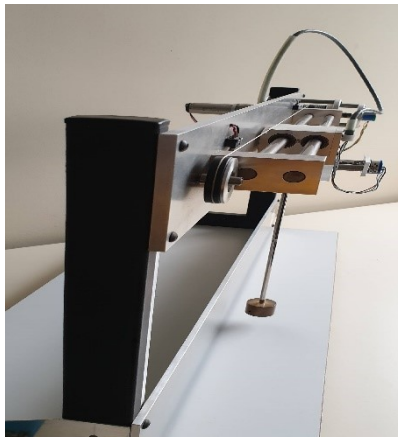


Figure 3: Maquette support de l'étude

The control input of the system is the voltage applied to the amplifier. The following measurements are available:

- A circular potentiometer gives a voltage v_θ proportional to the angular position at the output of the gearbox,
- A second potentiometer provides a voltage v_φ proportional to the angular position of the pendulum rod,
- A resistance, in series with the motor, provides a voltage v_i proportional to the current in the motor,
- An electronic device provides a voltage v_{Ω_m} proportional to the angular speed of the motor drive shaft.

1.2 Parametric knowledge model

In order to establish a knowledge model, a first step consists of breaking down the systems into subsystems and describing the behavior of each subsystem by equations resulting from physical laws.

1.2.1 Motorization system: amplifier / dc-motor-gearbox

The control voltage of the dc-motor is delivered by a power amplifier, which is modeled by a voltage gain associated with an output resistance (Figure 4). The following two equations are obtained:

- electrical equation $Au_c(t) = Ri_m(t) + L\frac{di_m(t)}{dt} + K_e\Omega_m(t)$
- mechanical equation $J\frac{d\Omega_m(t)}{dt} = K_c i_m(t) - \gamma_p(t) - \gamma_r(t) - f\Omega_m(t)$

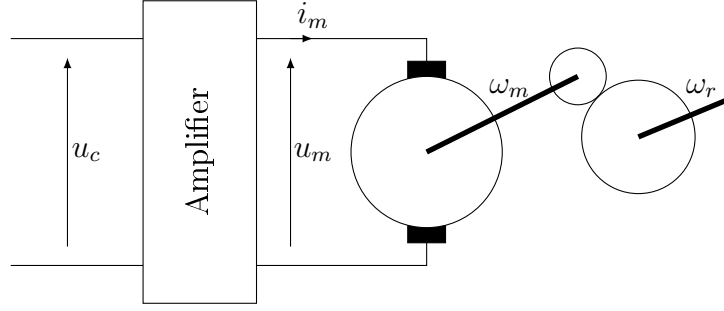


Figure 4: Physical model of the motorization system

with:

- $u_c(t)$: input voltage of the amplifier (in Volts);
- $u_m(t)$: DC current motor voltage (in Volts);
- $i_m(t)$: inductor current in the DC motor (in A);
- A : voltage gain of the amplifier;
- $R_m = R_s + R_{mot}$: total resistance (in ohm), where R_s is the output resistance of the amplifier and R_{mot} is the resistance of the DC motor inductor;
- L_m : inductance (in H) of the DC motor inductor;
- K_c : torque constant of the DC motor;
- ω_m : angular speed of the motor (in rad/s);
- γ_p : sum of the torques disturbances (except for the reaction torque created by the pendulum movement), brought back on the motor drive shaft (in N.m);
- γ_r : reaction torque (in N.m) created by the movement of the pendulum, brought back on the motor axis;
- $J = J_m + J_{rp} + (m_c + m_p) \left(\frac{r}{N}\right)^2$: total inertia (in $kg.m^2$) brought back on the motor axis, where J_m and J_{rp} are respectively the inertia of the unloaded motor and the inertia of the gearbox-pulley, and m_c et m_p are cart mass (in kg) and pendulum mass (in kg). N the gearbox ratio such that $N\Omega_r = \Omega_m$ and r the radius of the pulley (in m).
- f : viscous friction coefficient (in N.m.s).

1.2.2 Cart / pendulum

This is the set composed by the cart and the pendulum and shown in Figure 5.

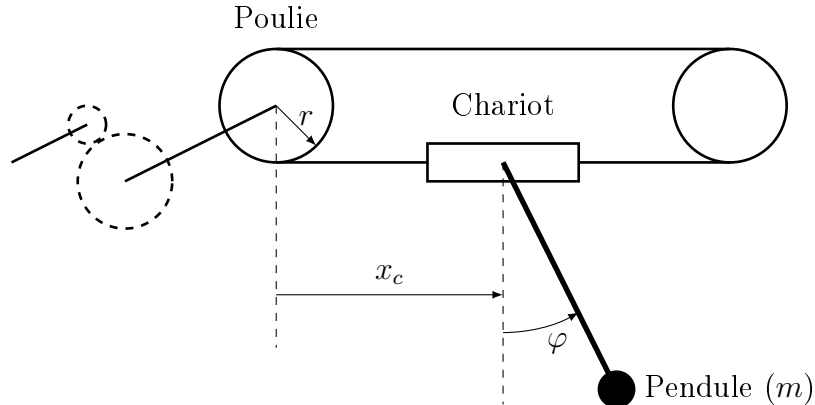


Figure 5: Cart - pendulum

The dynamical equations are:

- For the cart:

$$\frac{d\theta(t)}{dt} = \omega_r$$

whit $x_c(t) = r\theta(t)$ where r is the radius of the pulley, x_c is the position of the cart and θ the angle of rotation of the pulley.

- For the pendulum:

$$\cos \varphi(t) \frac{d^2 x_c(t)}{dt^2} + l \frac{d^2 \varphi(t)}{dt^2} = -g \sin \varphi(t) - a \frac{d\varphi(t)}{dt}$$

avec

- φ angle of the pendulum about the vertical (in rad);
- l length of the pendulum (in m);
- g acceleration of gravity (in m/s^2);
- a viscous friction coefficient (in m/s).

1.2.3 Sensors

The four sensors supply a voltage proportional to the angles θ and φ , to the motor current $i_m(t)$ and, to the motor angular speed $\Omega_m(t)$

$$v_\theta(t) = k_\theta \theta(t); \quad v_\phi(t) = k_\phi \varphi(t); \quad v_i(t) = k_i i(t); \quad v_\Omega(t) = k_\Omega \Omega_m(t)$$

1.3 Real-time control interface and numerical parameter values

- **Connection of converters**

- **CAN**

- * Voie 1: motor current measurement $v_i(t)$
- * Voie 2: angular speed measurement $v_{\Omega_m}(t)$
- * Voie 3: angular motor position measurement $v_{\theta}(t)$
- * Voie 4: angular deviation of the pendulum $v_\phi(t)$

- **CNA**

- * Voie 1: input $u_c(t)$ of the amplifier A
- * Voie 2: input setpoint $v_{xref}(t)$ (or display in the oscilloscope)

- **Sensor gains**

- Current measurement i_m : $k_i = 5 \text{ V} \cdot \text{A}^{-1}$
- Angular motor speed Ω_m : $k_\Omega = 0,02 \text{ Vs}$
- Measurement of the angle θ : $k_\theta = 0,875 \text{ V} \cdot \text{rad}^{-1}$
- Measurement of the angle ϕ : $k_\phi = 4,77 \text{ V} \cdot \text{rad}^{-1}$

- **Dc motor characteristics**

- Amplifier voltage gain: $A = 1$
- Ratio of the gearbox: $N = 17$
- Radius of the pulley: $r = 0,022 \text{ m}$
- Torque constant: $K_c(N\text{m/A})$ (depends on K_v and τ_m determined experimentally)
- Back EMF constant: $K_e(\text{Vs/rad})$ (We consider $K_e = K_c$)
- Motor armature inductance: $L_m = 0,2 \text{ mH}$
- Resistance of the motor inductor: $R_{mot} = 1,9 \Omega$
- Total resistance: $R_m = 2,25 \Omega$
- Moment of inertia of the unloaded motor: $J_m = 2 \cdot 10^{-6} \text{ kg} \cdot \text{m}^2$

- Moment of inertia of the loaded motor: $J = 3,9 \cdot 10^{-6} \text{ kg} \cdot \text{m}^2$
 - Friction coefficient: $f = 7,25 \cdot 10^{-5} \text{ Nm} \cdot \text{s}$
 - Mechanical time constant of the unloaded motor: $\tau_m = 15 \text{ ms}$
- **Cart characteristics for information only**
- Cart mass: $m_c = 0,5 \text{ kg}$
 - Load mass: $m = 0,1 \text{ kg}$

Part I

First session

2 Identification

2.1 Linear model and block diagram

Activity 1. Assuming the pendulum remains about the equilibrium point:

- Determine an approximate model in the form of linear differential equations.
- From the Laplace transforms of the obtained equations, establish the complete block diagram of the system, using all the variables describing its behavior.
- Assuming that the motor inductance can be neglected, $L \approx 0$, give the analytic expression of the following transfer functions (s being the Laplace variable):

$$H_I(s) = \frac{I(s)}{U_c(s)} \quad H_\Omega(s) = \frac{\Omega_m(s)}{U_c(s)}$$

Put $H_\Omega(s)$ in the form $\frac{\Omega_m(s)}{U_c(s)} = A \frac{K_v}{1 + \tau_m s}$, where K_v is called the speed constant of the motor and τ_m its mechanical time constant (with load).

The identification of the parameters of the manipulated models, sections 2.2 and 2.3, can be carried out by exploiting oscilloscope readings or by exploiting measurements made with the control diagram ("**MesuresEtTestES.slx**") provided in the Matlab/Simulink environment. This diagram is used to control the electrical machine and to save in the workspace in the form of column matrices the samplings of the following signals:

- input voltage (variable "**cmd**"),
- cart position (variable "**Xc**"),
- angular speed of the motor (variable "**Omega_m**"),
- angular position (variable "**Phi**"),
- motor current (variable "**im**"),
- and a vector time corresponding to the sampling time instants (variable "**temps**").

The file "**DataPontRoulant.m**" allows to initialize a certain number of parameters necessary to the real time control like the sampling time T_e for example. This command file must be executed in advance (once) before the first realization of the command in real time.

2.2 Identification of the motor parameters

The amplifier offset will be adjusted to cancel any dc components at the motor input and thus avoid any drift of the cart.

Activity 2. Read the time response, to a step voltage of peak to peak amplitude of 6 V and frequency of 0.5 Hz at the amplifier input, of the current measurement $i_m(t)$ and of the speed measurement $\Omega_m(t)$. The file "**MesuresEtTestES.xls**" can be used for this activity. Once the cabling defined in ?? is done, the real time task can be executed with the

button .

Activity 3. By using these records, determine k_v and τ .

Activity 4. Validation of identified parameters : compare the measured response with that obtained with the identified model. For this comparison the file "**Validation_KvTaum.m**" is provided.

2.3 Identification of the pendulum parameters

It is proposed to identify the characteristic parameters of the pendulum movement from the recording of the angular evolution of the pendulum following a release from an initial position, without initial speed, the cart being blocked in an arbitrary position. The same file "**Mesure-sEtTestES.slx**" can be used for this recording by changing the amplitude of the control voltage to zero.

Activity 5. Show that the linear differential equation describing the motion of the pendulum in these conditions, under the assumption that $\sin \varphi \approx \varphi$ is

$$\ddot{\varphi}(t) + 2\xi\omega_0\dot{\varphi}(t) + \omega_0^2\varphi(t) = 0$$

and precise the relation between the parameters ξ and ω_0 and the physical parameters of the pendulum.

By solving this equation, one may show that the evolution of φ presents a pseudo-period $T = \frac{2\pi}{\omega_0\sqrt{1-\xi^2}}$, and successive extrema such that

$$\ln \left(\frac{|\varphi(0)|}{|\varphi(\frac{kT}{2})|} \right) = k\eta$$

with $\eta = \frac{\pi\xi}{\sqrt{1-\xi^2}}$.

Activity 6. From the recording of a pendulum time response

- determine the pseudo period T ;
- from two successive extrema of the time response, calculate the value of the parameter ξ , deduce the value of the parameter ω_0 , then those of the parameters l and a involved in equation describing the pendulum movement;
- using a Matlab functionalities, check the system responses with the parameters determined above.

Activity 7. Validation of identified a and l parameters

- write the state space representation corresponding to the linearisation of the equation with φ as output;
- using a Matlab functionalities, check the system responses with the parameters determined above. The Matlab function `initial` can be useful in this endeavour.¹ for this comparison the file "**Validation_param_pend.m**" is provided.

¹It should be noted that for a vector `temps` and a constant `a`, the command `index = temps>a; temps_tronc = temps(index);` generates a vector `temps_tronc` composed of the elements of `temps` having a value larger than `a`.

2.4 Analysis of a control using a simplified model

In order to highlight the behavior of the pendular motion of the overhead crane and the induced problems we propose to realize the control of the cart alone by neglecting the influence of the pendulum.

For this purpose, a simple model is used by considering a simplified model to establish the control law, where the state space vector is composed of the speed ω_m and the position of the motor θ_m , the input signal is $u_c(t)$.

The specifications impose the following constraints:

- in steady state, the position of the cart must be regulated to a reference value x_{ref} without error;
- from a dynamic point of view, we want a time response $t_m \approx 1$ s without overshoot.

Activity 8. Establish a state representation of the simplified model by considering as state variables x_c and ω_m , that is $z = [x_c \ \omega_m]^\top$, and $u_c(t)$ as input signal. We will consider that all the state space variables are measurable and know.

Determine a the state space representation of the augmented model, including in the state space vector a new state space variable $z_i(t) = \int(x_c - x_{ref})d\tau$, that is $x = [x_c \ \Omega_m \ z_i]^\top$.

Activity 9. Design of the state feedback control

- determine a feedback control law $u_c(t) = -K_w x(t)$ allowing closed loop dynamics given by the eigenvalues:
 $[-\xi\omega_0 - i\omega_0\sqrt{1-\xi^2}; -\xi\omega_0 + i\omega_0\sqrt{1-\xi^2}]$ avec $\omega_0 = 3$ et $\xi = 0.9$;
- using Matlab's functions and the simplified model, simulate the behaviour at a reference step x_{ref} of amplitude 30cm. Display the output signal $x_c(t)$, the reference signal x_{ref} and the control signal $u_c(t)$. Check that the control signal respects the $\pm 10V$ constraint. Are the specifications met?
- comment on the behaviour obtained with respect to the eigenvalues of the closed loop.

Activity 10. Verification of the control law in the non-linear simulation

Before testing the synthesised control law on the real system, it must be verified in a simulation file containing the non-linear model of the platform, the sensors and the actuators. This file is named "**SimulateurNL_Pont_Cmd_Kw.slx**". To use this file, you need to enter the values identified by K_v , τ_m , a and l in the "SIMULATEUR" block. A double click in the block shown in figure 6 displays the parameter block where the identified values are entered.

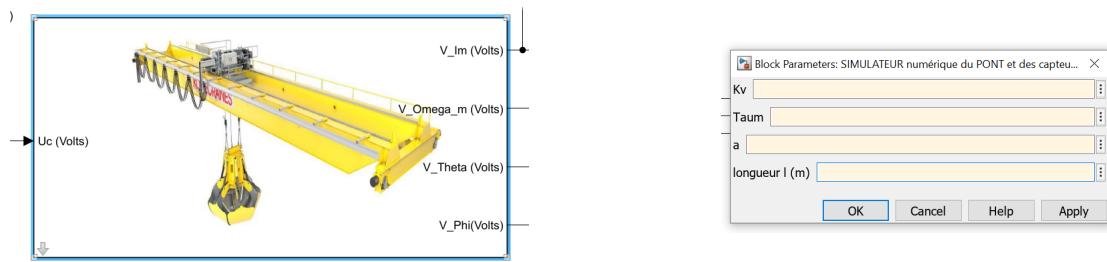


Figure 6: Non Linear simulator of PONT

- determine the closed loop responses (cart position $x_c(t)$, pendulum $\varphi(t)$ and motor input voltage $u_c(t)$) in response to a step reference of amplitude $x_c = 15$ cm.

2.5 Validation expérimentale

For the real time experimental implementation of the closed loop control of the cart:

- the real time control file "**ImplantationCmd_Kw.slx**" is provided;
- the feedback matrix is noted $K_w = [k_1 \quad k_2 \quad k_3]$ where k_3 corresponds to the gain of the integral action;
- the real time numerical realization of the control and the sampling of the various quantities is carried out using the sampling period noted $T_e = 0.01s$;
- in addition to the quantities recorded in the case of open loop control, a column vector records the samples of the setpoint (variable "**Xc_ref**");
- if necessary, the numerical assignment of the various parameters can be carried out beforehand, for example by completing the command file "**DataPontRoulant.m**".

Activity 11. Experimental validation:

- realise the regulation for step reference signals of amplitude $X_0 = 15$ cm.
- Compare the performances obtained experimentally with those obtained in simulation. Analyse in particular the influence of the sampling time on the closed loop performances.

Activity 12. Collect the different results (simulation and experimental data) and conclude on the validity of the identified model and on the performance of the regulation.

Part II

Preparatory work between the two sessions: Synthesis of a control law

In this work, to be done at home, we want to determine the parameters of the control law allowing the displacement of the overhead crane so as to minimize the pendulum movement while ensuring the following specifications:

- in steady state, the position of the cart must be regulated to a reference value x_{ref} without error;
- from a dynamic point of view we want:
 - a time response $t_m \approx 1$ s,
 - no static error and an overshoot $D < 5\%$.

The control law obtained in this preparatory work will be validated experimentally during the second session of laboratory work. The absence of rigorous preparation of the elements necessary for the second session will be penalizing. Thus, in this preparatory work, we will first elaborate the state feedback control law by exploiting the LQ approach and then we will determine an observer allowing to obtain an estimate of the state from the position measurements only.

Validation of control laws

To carry out this preparation the models necessary for the continuation of this work are partially provided, in the form of a state representation, in the file "**ModelesEtParametresPont.m**" to define the parameters of the model and the state space representation of the overhead crane.

The validation of the performance of the control laws and the convergence of the observer will be done in simulation:

- using the linear models obtained by linearisation;
- for the synthesis, the model obtained after the first session will be used and the weighting matrices can be adjusted according to the answers to an initial condition;
- for the validation in simulation regarding the response to a step reference signal in the displacement, the models "**SimulateurNL_Pont_Cmd_LQ.slx**" and "**SimulateurNL_Pont_Cmd_LQI.slx**" can be used. These files correspond respectively to the feedback control without and with an integral action.
- the real time implementation, during the 2nd session, will be done numerically with a sampling time T_{e1} .

Steps in the synthesis of the control law

LQ synthesis of the state feedback control

- Control law without integral action
 - Specify the state-space representation of the simplified model by considering $u_c(t)$ as input control signal. The state space vector is $z(t) = [x_c \quad \Omega_m \quad \varphi \quad \dot{\varphi}]^T$ and the

measurement vector is $y(t) = [x_c \quad \varphi]^\top$.

$$\begin{cases} \dot{z}(t) = A_p z(t) + B_p u_c(t) \\ y(t) = C_p z(t) \end{cases}$$

- Using the LQ approach, determining state feedback control

$$u_c(t) = -K_{LQ} z(t) + M_{LQ} x_{ref}$$

to check the expected specifications.

Matlab functions will be used to synthesise the LQ controller. The command file **"DataPontRoulant.m"** can be completed for the synthesis of the control law. The final validation in simulation could be done with the simulation model **"SimulateurNL_Pont_Cmd_LQ.slx"**.

Note : The synthesis of the LQ controller minimises the following criterion:

$$J = \int_0^\infty (z^\top Q z + u_c^\top R u_c) dt \quad (1)$$

where the weighting matrices Q and R need to be determined in order to obtain a control system that meets the specifications. In order to find the Q and R matrices, one possible approach is to normalise the state variables with respect to their maximum values and consider only the outputs to be controlled. Thus, $\tilde{x}_i = \frac{x_i}{x_{imax}}$ and with $\tilde{y} = C\tilde{x}$ and expressing the criterion with the normalised variables, we have :

$$J = \int_0^\infty (\tilde{y}^\top \tilde{Q} \tilde{y} + \tilde{u}_c^\top \tilde{R} \tilde{u}_c) dt = \int_0^\infty (\tilde{z}^\top C^\top \tilde{Q} C \tilde{z} + \tilde{u}_c^\top \tilde{R} \tilde{u}_c) dt \quad (2)$$

Unitary diagonal matrices for the $\tilde{Q}$ and $\tilde{R}$ usually provide a good starting point for iterative adjustment. For slower overall behaviour we increase $\tilde{R}$ and for faster overall behaviour we decrease $\tilde{R}$. If we want to make a particular state variable faster or slower, we respectively decrease or increase the corresponding $\tilde{Q}_i$ weighting. We then return to the criterion in its initial form (1) using matrices with normalisation terms.

$$\begin{aligned} \tilde{z}^\top C^\top \tilde{Q} C \tilde{z} &= z^\top \underbrace{C^\top Q_n \tilde{Q} Q_n C}_{\tilde{Q}} \tilde{z} \\ \tilde{u}_c^\top \tilde{R} \tilde{u}_c &= u_c^\top \underbrace{R_n \tilde{R} R_n}_{\tilde{R}} u_c \end{aligned}$$

With:

$$Q_n = \begin{bmatrix} \frac{1}{y_{1\max}} & 0 & 0 \\ 0 & \ddots & \vdots \\ 0 & \dots & \frac{1}{y_{n\max}} \end{bmatrix}$$

$$R_n = \begin{bmatrix} \frac{1}{u_{1\max}} & 0 & \vdots \\ 0 & \ddots & \vdots \\ 0 & \dots & \frac{1}{u_{m\max}} \end{bmatrix}$$

In the case of the Pont platform, an initial choice is to start with $\tilde{y}(t) = [\tilde{x}_c \quad \tilde{\varphi}]^\top$ with, $\tilde{Q} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ et $\tilde{R} = 1$. The following maximum values will be considered:

$$\begin{aligned} x_{c\max} &= 0,5m, \\ \Omega_{m\max} &= 500rad/s, \\ \varphi_{\max} &= \pi/4rad, \\ \dot{\varphi}_{\max} &= 1rad/s. \end{aligned}$$

- Control law with integral action

- Complete the state space vector with an integral action $z_i(t) = \int (x_c - x_{ref}) d\tau$:

$$x(t) = [x_c \quad \omega_m \quad \varphi \quad \dot{\varphi} \quad z_i]^\top$$

- Using the LQ approach, determine the state feedback $u(t) = -K_{LQ} x_a(t)$ to verify the expected specifications. In a similar way to the control without integral action, the Matlab functions will be used to synthesise the LQ controller. The command file **"DataPontRoulant.m"** can be completed to synthesise the control law. The final simulation validation can be carried out using the simulation model **"SimulateurNL_Pont_Cmd_LQI.slx"**. For the choice of weights, consider $\tilde{y}(t) = [\tilde{x}_c \quad \tilde{\varphi} \quad z_i]^\top$ and $z_{imax} = 0.15$.

- Collect the results of the two control structures and compare the obtained performance.

Synthèse d'un observateur

Using the model of the system (control input $u_c(t)$, state space vector $z = [x_c \quad \Omega_m \quad \varphi \quad \dot{\varphi}]^\top$ and measurement vector $y(t) = [x_c \quad \varphi]^\top$, determine an observer allowing to obtain an estimate $\hat{z}(t)$ of the state space vector.

The command file "**DataPontRoulant.m**" can be completed for the observer's synthesis. The validation in simulation of the convergence of the observer can be done by using the simulation model "**SimulateurNL_Pont_Cmd_LQGI.slx**" corresponding to a state feedback control completed by an observer.

Part III

Experimental realization of the control law

In this part, the control laws which are the subject of the preparatory work will be experimentally implemented and the obtained performance will be evaluated. The preparatory work must have been done.

3 State feedback control law without observer

3.1 General elements of the study to be realized.

First, it is assumed that the set of state space variables is accessible to the measurement and know:

- for the real time realization a sampling time $T_{e1} = 1$ ms will be used;
- only the angular positions being accessible to the measurement, the speeds will be calculated by numerical derivation.

3.2 Control scheme

For the real time impementation of the control law:

- the Matlab command file "**DataPontRoulant.m**" allows to initialize the parameters of the control law;
- the files "**ImplantationCmd_LQ.slx**" and "**ImplantationCmd_LQI.slx**" will be used to the real time realization without and with an integral action respectively.

These files (provided) memorize in a colon matrix the following signals:

- input control voltage (variable "**cmd**");
- position of the cart (variable "**Xc**");
- cart position reference (variable "**Xc_ref**");
- angular speed of the motor shaft (variable "**Omega_m**");
- angular position of the pendulum (variable "**Phi**");
- angular speed of the pendulum (variable "**vit_Phi**");
- and a time vector corresponding to the sampling instants (variable "**temps**").

Note in these files that:

- the speeds are obtained by derivation of the angular positions;
- the state feedback matrix is named K_{lqi} in the case of the control law with integral action and it is decomposed as $K_{lqi} = [k_1 \ k_2 \ k_3 \ k_4 \ k_5]$ where k_5 is the gain of the integral action;
- the integral action is realized in discrete time because of the numerical realization of the control. The choice of the sampling period must be made accordingly to the dynamics of the closed loop system.

3.3 Work to be done

In all cases, in addition to the signals of evolution of the positions of the cart and the pendulum, it will be relevant to record the control signal (motor voltage) in order to analyze the performances of the various envisaged control laws.

3.3.1 Control without integral action

For this part the file "**ImplantationCmd_LQ.slx**" can be used to the real time implementation. The parameters of the control law will be those determined during the preparatory work.

Activity 13. Check if the control diagram provided allows to regulate the position of the carriage to a constant reference position x_{ref} . Assign or modify, if necessary, the parameters of the control law with respect to those predetermined and validate the results obtained experimentally.

Activity 14. Analyze the behavior of the system for references of different amplitudes (it will be possible, for example, to use a square setpoint signal). Check if the requirements of the specifications are ensured both from the point of view of dynamic behavior and static error.

3.3.2 Control with integral action

Activity 15. Repeat the study of the activities 13 to 14 with the control law with integral action. Use the file "**ImplantationCmd_LQI.slx**".

Activity 16. Compare the performance of the two control laws.

4 State feedback and observer control

In this section, the control law is implemented using an observer that estimates the entire state vector from the measurement $y(t) = [x_c \quad \varphi]^\top$. For this we will use the observer previously designed in the preparatory work and implemented numerically by discretizing the continuous time representation of the observer.

For the real time implementation of the control law the file "**ImplantationCommandePontEtatAvecIntObs.slx**" and the command file "**DataPontRoulant.m**" allowing to initialise the control law parameters are provided.

The observer is realised in discrete form by performing the following approximation (which corresponds to the Euler transform):

$$\begin{cases} \hat{x}(t) = A\hat{x}(t) + Bu(t) + L(y(t) - \hat{y}(t)) \\ \hat{y}(t) = C\hat{x}(t) \end{cases}$$

$$\frac{d\hat{X}(t)}{dt} \approx \frac{\hat{X}((k+1)T_e) - \hat{X}(kT_e)}{T_e}$$

from which we derive the discrete equation of state:

$$\hat{X}((k+1)T_e) = [I + (A - LC)T_e] \hat{X}(kT_e) + BT_e u(kT_e) + LT_e y(kT_e)$$

and the control equation :

$$u(kT_e) = -K \hat{X}(kT_e)$$

The implementation, and validation, of the control law with observer will be done by following the following activities.

4.1 Time performance analysis

Activity 17. The file "**ImplantationCmd_LQGI.slx**" is used to loop through the state feedback control using the quantities obtained from the observer. Record and compare the performance obtained with that observed previously.

Activity 18. Analyze the performance of the control law when the sampling period varies $T_e \in [1 ; 20]$ ms.

4.2 Analysis of stability margins

This part, which will allow us to conclude on the performances of the different control laws, can be done at home. In order to establish a comparative report, as complete as possible, it is necessary in this part to analyze the margins of stability of the control laws without and with observer.

Activity 19. In this activity the properties of the control with integral action and without observer are considered. Using the open loop state space representation in the form:

$$\begin{cases} x(t) = Ax(t) + Bu(t) \\ r(t) = Kx(t) \end{cases}$$

where K is the state space feedback matrix, determine the gain and phase stability margins using the frequency responses of the function $\frac{R(s)}{U(s)}$.

Activity 20. In this part, the effect of the observer on the stability margins is taken into account. For this analysis it will be necessary, in advance, to establish the state representation of the system augmented by the observer. For this we use the notation $x_a(t) = [x^\top \int(x_c - x_{ref})dt \quad \hat{x}^\top]^\top$ for the augmented state vector composed of the process state variables, the integral action term and those of the observer.

(a) Determine the open loop state representation of the system with observer

$$\begin{cases} \dot{x}_a(t) = A_a x_a(t) + B_a u(t) + B'_a x_{ref} \\ r(t) = K' x_a(t) \end{cases}$$

The matrices A_a and B_a , will be obtained according to the representations of the process and the observer, that is, according to the matrices A, B, C, L , the feedback matrix K' will be determined according to the components of the state feedback matrix $K = [K_1 \ K_2]$.

(b) Plot the frequency response of the function $\frac{R(s)}{U(s)}$ and then determine the stability margins using this model.

5 Conclusion and comparative analysis

Activity 21. Summarize the performances of the different control laws considered and propose a conclusion on the whole study.

